Module Test 6 Preparation Guide

DA3452 · Speech and Language Technology
Portion: Blocks 3, 4 and 5 · 4 November 2026

Course pages

Block 3: Preference learning and RL foundations
Block 4: Policy gradients
Block 5: Reference-policy regularisation, PPO and DPO

Rework the numerical examples covered in class.
The numerical examples in the class slides and Block 3–5 HTML are part of your preparation. Work through them on paper, explain each step, and repeat them with changed values. Reading the theory or memorising the final answers is not sufficient. Use the revision checklist on page 6.

Use the course narratives, slides and videos first. The textbook and focused readings below support the same topics. A reference to a chapter or paper does not add its remaining contents to the test portion.

Topic and reading map

Topic	Focused reading	Course location
Preference data and reward models	J&M §§8.3–8.3.3, pp. 215–220; Eqs. 8.2–8.4; Figs. 8.9–8.10.	Block 3 ML01–ML02
RL formulation and expected reward	J&M §8.4, pp. 219–220, Eq. 8.5. S&B Ch. 2 opening and §3.1.	Block 3 ML03–ML04; Block 4 ML01
Policy gradient, REINFORCE and baselines	S&B §§13.1, 13.3–13.4, pp. 321–331. J&M supplies the objective, not the full derivation.	Block 4 ML01–ML03
Reference policy and KL penalty	J&M §8.4, p. 221, Eqs. 8.6–8.7. Course calculations and gradient.	Block 5 ML01
PPO and sampled-batch reuse	PPO paper [R3], §3, Eqs. 6–7 and Fig. 1; §5 for the update loop. J&M §8.4.1 is TBD.	Block 5 ML02
DPO	J&M §8.4.2, pp. 222–223, Eqs. 8.8–8.13 and unnumbered losses; DPO paper [R4], §4.	Block 5 ML03

Reference convention. J&M is the supplied Speech and Language Processing draft of 19 August 2026, used in these blocks and in the Test 5 guide. Page numbers are printed numbers; PDF viewer page = printed page + 8. If you use another draft, match section titles because numbering can differ. S&B is Sutton and Barto, 2nd edition (2018). [R1]–[R5] are on page 7.

1. Block 3: preferences and the RL formulation

Preference data: what the labels mean

• Explain why comparisons can be easier to collect than expert-written target responses. Distinguish demonstration data from preference data and identify the prompt, chosen response and rejected response.

• Know the data sources discussed: human comparisons, rankings, votes and AI feedback. A chosen response is preferred within the shown pair; the label does not certify factual correctness or identify the best possible response.

• Convert a full ranking of N responses into $N(N-1)/2$ pairs. A best-only annotation gives $N-1$ pairs. Explain which comparisons the best-only annotation leaves unknown.

Read: J&M §8.3 and §8.3.1, pp. 215–217. Paper context: [R2].

Reward scores, Bradley–Terry probability and loss

The reward model $r_\phi(x,o)$ gives one scalar score to a prompt–response pair. The shared scorer evaluates both responses; score differences determine preference probabilities.

$$\delta = r_\phi(x,o_w) - r_\phi(x,o_l),\ P(o_w \succ o_l \mid x) = \sigma(\delta)$$
$$L_{pair} = -\ln \sigma(\delta),\ dL_{pair}/d\delta = \sigma(\delta) - 1$$

• Compute the score gap, preference probability, pair loss and direction of the score update. Explain why adding a common constant to all scores for one prompt leaves its preference probabilities unchanged.

• Rework the disagreement example: three annotations prefer A and one prefers B. With a freely fitted gap, the optimum is $\sigma(\delta) = 3/4$ and $\delta = \ln 3$. A shared model fitted across many pairs may need to compromise.

• Explain the scalar output head, training on comparison labels, generalisation to unseen responses, and best-of-N selection. Selecting the highest-scoring candidate does not itself update the generator.

Read: J&M §§8.3.2–8.3.3, pp. 218–220; Eqs. 8.2–8.4 and Fig. 8.10.

Why reinforcement learning?

• Distinguish instructive feedback from evaluative feedback. A reward score for a completed response does not supply a correct target token at each step.

• Map the generator to an agent: state = prompt plus generated prefix; action = next token; transition = append that token; policy = token distribution. EOS terminates the response. In the terminal-reward example, earlier rewards are zero and the completed response receives the learned score.

• Explain why the fixed reward model can guide a changing policy, and why a learned reward is a surrogate for the qualities that annotators value.

Read: J&M §8.4 through Eq. 8.5, pp. 219–220; S&B Ch. 2 opening and §3.1.

> **Numerical work to revisit**
> Pair counts; the scores 2.0 and 0.5 and their reversed ordering; the 3-to-1 disagreement loss; and the token-by-token episode with a terminal reward.

2. Block 4: policy gradients and baselines

Complete-response probability and expected reward

$$\pi_\theta(o \mid x) = \prod_{t=1\ldots T} \pi_\theta(o_t \mid x, o_{<t})$$
$$J_x(\theta) = \sum_o \pi_\theta(o \mid x)\, r_\phi(x,o)$$

Include EOS in the response probability. Distinguish an episode reward, a finite-sample average, the expectation for one prompt, and an average across a stated prompt distribution. Prompt weights are equal only for a uniform distribution.

In the two-response example, $p = \sigma(z)$, rewards are 2 and -1, and $J_x = 3p - 1$. At $p = 0.75$, $J_x = 1.25$. This generation probability is a different event from the preference probability in Block 3.

Deriving and estimating the gradient

$$\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \ln \pi_\theta$$
$$\nabla_\theta J = E[r_\phi(x,o) \nabla_\theta \ln \pi_\theta(o \mid x)]$$

• Follow the derivation from a probability-weighted sum to an expectation. The generator changes the sampling probabilities; the reward-model parameters stay fixed. Do not differentiate through a discrete sampled token as though it were a continuous output.

• Use $\ln \pi(o|x) = \sum_t \ln \pi(o_t|x,o_{<t})$. Include response tokens and EOS; exclude prompt and padding targets. Compute $dJ_x/dz = 3p(1-p)$, the single-sample estimate and a batch mean.

• Rework the update $z \leftarrow z + \alpha g$, then recompute p and J_x. A finite batch need not equal the exact gradient. For N independent samples from a fixed policy, the mean estimator has variance $\mathrm{Var}(g)/N$.

Reward offsets and baselines

$$g = [r_\phi(x,o) - b(x)] \nabla_\theta \ln \pi_\theta(o \mid x)$$
$$E[b(x) \nabla_\theta \ln \pi_\theta(o \mid x)] = 0$$

• Adding a common reward offset changes the expected objective by a constant and leaves its gradient unchanged, but can greatly change single-sample variance.

• For the taught estimator, the baseline may depend on the prompt but not the sampled response. Explain the zero-expectation identity. A mean-reward baseline removes a common offset; it does not always reduce variance relative to using no baseline.

• Trace the loop: collect a fresh batch, score responses, predict baselines, form detached advantages, update the policy, then fit the baseline by squared error. Use the baseline prediction from before fitting on that batch.

Read: J&M §3.1, p. 70, Eqs. 3.3–3.4; §8.4, p. 220, Eq. 8.5. S&B §13.1 and §§13.3–13.4; original REINFORCE reference [R5].

> **Numerical work to revisit**
> The cat policy; exact and sampled gradients; one update with $\alpha = 0.1$; sample variance and batch size; the +10 reward offset; the mean baseline and its variance comparison.

3. Block 5: reference policy and PPO

Reference-policy regularisation

The frozen reference π_{ref} is copied from the SFT model. It penalises departure from that distribution. The reward model remains frozen during policy updates. A KL penalty can limit exploitation of a scorer error; it does not repair the scorer or guarantee correctness.

$$\ell_\theta(x,o) = \ln[\pi_\theta(o \mid x) / \pi_{ref}(o \mid x)]$$
$$D_{KL}(\pi_\theta \,||\, \pi_{ref}) = E_{o\sim\pi\theta}[\ell_\theta(x,o)]$$
$$J_\beta = E[r_\phi(x,o) - \beta\ell_\theta(x,o)]$$

• Compute both response log-ratios, their current-policy-weighted KL, the adjusted rewards and the regularised objective. An individual log-ratio can be negative; the full KL is nonnegative.

• Explain the role of β and the scale of the rewards. The penalty changes the objective; the Block 4 baseline leaves its expected gradient unchanged.

• Follow the taught gradient argument: the extra term is $-\beta E[\nabla \ln \pi_\theta] = 0$. The sampled adjusted reward is held constant during the score-function policy update. Revisit the full RLHF loop with the reference and baseline.

Read: J&M §8.4, p. 221, Eqs. 8.6–8.7; Block 5 ML01 for the calculation and gradient.

PPO: reusing a batch collected under an earlier policy

π_{old} is the policy that generated the rollout batch. It stays fixed over the updates on that batch and is refreshed for a new rollout. π_{ref} stays fixed across training. Their roles and denominators differ.

$$\rho = \pi_\theta(\text{action} \mid \text{context}) / \pi_{old}(\text{action} \mid \text{context})$$
$$L^{CLIP} = \text{mean}[\min(\rho\hat{A}, \text{clip}(\rho, 1-\varepsilon, 1+\varepsilon)\hat{A})]$$

• Explain why reusing old samples in an unchanged on-policy estimate gives the wrong weighting after the policy moves. Compute the probability ratio and trace the reweighted cat batch. The cat example has only one uncertain token, so its action ratio equals its response ratio.

• For positive and negative advantages, calculate both products before taking the minimum. Explain which direction becomes flat. Clipping removes further incentive in that direction; it does not force every ratio to remain inside the interval.

• Distinguish PPO objective clipping from gradient-norm clipping. Trace repeated minibatch updates with stored rollout probabilities and fixed advantages, followed by a fresh rollout.

Read: Block 5 ML02 and [R3] §3, Eqs. 6–7 and Fig. 1; §5 for the loop. S&B §5.5, Eq. 5.3 supports the importance-sampling idea. J&M §8.4.1 is marked TBD in the supplied draft.

Numerical work to revisit
The KL and adjusted-reward table; the effect of β and reference odds; the stored 3:1 cat batch; reweighted reward and gradient; and the clipping table with $\varepsilon = 0.2$.

4. Block 5: direct preference optimisation

From the KL-regularised optimum to a preference loss

For a fixed prompt and reward, the optimum over response distributions reweights the reference by the exponentiated reward. Follow the normalisation and the role of the partition function Z(x).

$$\pi^*(o \mid x) = \pi_{ref}(o \mid x) \exp[r(x,o)/\beta] / Z(x)$$
$$r(x,o) = \beta \ln[\pi^*(o \mid x)/\pi_{ref}(o \mid x)] + \beta \ln Z(x)$$

Substitute the two reward expressions into Bradley–Terry. Because both responses share the same prompt, the $\beta \ln Z(x)$ terms cancel. Replacing π^* by the trainable π_θ gives the DPO margin and loss.

$$m_\theta = \beta\{\ln[\pi_\theta(o_w|x)/\pi_{ref}(o_w|x)] - \ln[\pi_\theta(o_l|x)/\pi_{ref}(o_l|x)]\}$$
$$L_{DPO} = -\ln \sigma(m_\theta)$$

• Compute response log-probabilities by summing token log-probabilities, including EOS and excluding prompt and padding targets. Then compute both log-ratios, the margin, predicted preference probability and loss.

• Distinguish $\sigma(m_\theta)$, which predicts a preference label, from the probability that the generator produces a response. A preferred response can be more probable than its alternative and still have a negative DPO margin if its relative odds have fallen from the reference.

What the labels determine

• Rework the one-sided-label case and the mixed-label case. For a freely fitted pair with fraction q preferring the winner identity, the mean-loss minimum has $\sigma(m) = q$. Consistently one-sided labels can keep pushing the odds towards an extreme.

• Explain why the closed-form optimum for a specified reward is not automatically the minimum for an arbitrary finite preference dataset. The connection uses the preference model, comparison coverage, representability and optimisation assumptions taught in ML03.

• The DPO loss favours increasing the preferred-versus-rejected margin relative to the reference. With shared parameters, it does not guarantee that one individual response probability rises while the other falls. It also does not impose a hard KL budget.

Training procedure and comparison with PPO

Role	PPO-based RLHF	Standard offline DPO
Trainable policy	Updated from sampled rollout feedback	Updated from fixed preference pairs
Reference	Frozen; measures departure	Frozen; supplies reference log-probabilities
Other model roles	Frozen reward model; learned value/baseline model	No separately trained reward or value model
New responses in training	Collect new rollouts as the policy changes	No online generation required by the training loss

Read: J&M §8.4.2, pp. 222–223; Eqs. 8.8–8.13 and the unnumbered losses; [R4] §4, Eqs. 4–7. Reproduce the derivation steps shown in the course; the full appendix is optional.

5. Numerical revision checklist

Rework these examples from the class slides and HTML without looking at the solution. For each, write the governing expression, substitute the values, check the result and explain its meaning. Then change a probability, reward, label count or coefficient and repeat.

Location	Numerical work to practise
Block 3 ML01	Pair counts from full rankings and best-only labels; identify comparisons that are not supplied.
Block 3 ML02	Scores 2.0 and 0.5: $\delta = 1.5$, $\sigma(\delta) \approx 0.8176$, loss ≈ 0.2014. Reverse the scores and recompute.
Block 3 ML02	Three votes for A and one for B: write both directions of the loss, differentiate, obtain $p = 0.75$ and $\delta = \ln 3$.
Block 4 ML01	Multiply token probabilities through EOS; compute expected reward for one prompt and a weighted prompt collection; compare with a finite-sample mean.
Block 4 ML02	Cat example $p = \sigma(z)$, rewards 2 and -1: derive $J = 3p - 1$ and $dJ/dz = 3p(1 - p)$. At $p = 0.75$ the slope is 0.5625.
Block 4 ML03	Single-sample estimates 0.50 and 0.75; batch averages; $\alpha = 0.1$ update from $z = \ln 3$; recompute $p \approx 0.7604$ and $J \approx 1.2812$.
Block 4 ML03	Compute variance and the effect of N independent samples. Add 10 to both rewards; compare the mean gradient and variance before and after subtracting the mean baseline.
Block 5 ML01	Reference probabilities 0.75 and 0.25, $\beta = 2$: compute log-ratios, KL, adjusted rewards and $J\beta$ at $p = 0.75$, 0.90 and 0.99. Explain why higher raw reward can give a lower regularised objective.
Block 5 ML01 / ML03	Use optimal odds = reference odds × exp(reward gap/β). Change the reference probability or β; interpret the resulting response probability and scorer-error amplification.
Block 5 ML02	Stored batch sat, sat, sat, flew: $b = 1.25$; advantages 0.75 and -2.25. At current $p = 0.80$, compare unchanged and reweighted reward/gradient estimates.
Block 5 ML02	With $\varepsilon = 0.2$, compute both ratios and both surrogate products. Reproduce the clipping table at $p = 0.75$, 0.80, 0.90 and 0.95.
Block 5 ML03	With $\beta = 2$ and reference 0.75/0.25, compute the DPO margin, preference probability and loss at $p = 0.60$, 0.75 and 0.90. At the reference, the margin is 0 and loss is $\ln 2$.
Block 5 ML03	Compare one-sided labels with mixed labels. For a 3:1 label split, solve $\sigma(m) = 0.75$: the cat-policy optimum is $p \approx 0.839$, not 0.75. Explain why.

Conventions: use natural logarithms for these losses and log-ratios. State whether a derivative is with respect to p or the logit z. A sample average uses the observed counts; an exact expectation uses the policy probabilities. The numerical values above are checkpoints, not a substitute for the working.

6. Focused sources and final preparation

Read the indicated parts to clarify the course material. Full papers, their experimental catalogues and unlisted appendices are not additional test portions. References support the concepts and calculations already taught.

Textbooks

J&M. Jurafsky and Martin, Speech and Language Processing, supplied 19 August 2026 draft. Use the exact section boundaries on page 1. Official textbook website. The online draft may differ from the course PDF.

[R1] Sutton and Barto (2018). Reinforcement Learning: An Introduction, 2nd ed. Focused self-study: Ch. 2 opening (feedback), §3.1 (agent–environment interface), §13.1 (policy parameterisation), §13.3, pp. 326–328 (REINFORCE), §13.4, pp. 329–331 (baseline); §5.5, Eq. 5.3 for the ratio-weighting idea. Learn the applications and derivations shown in the blocks, not the entire chapters. Author's book page.

Papers with specific reading targets

[R2] Ouyang et al. (2022). Training Language Models to Follow Instructions with Human Feedback. Read Fig. 2 and §3.1 for the SFT → reward model → PPO pipeline; §3.5 for the reward-model and RL descriptions. Use as context for the model roles and preference collection already covered. Numerical settings and experimental results are not memorisation targets. arXiv:2203.02155.

[R3] Schulman et al. (2017). Proximal Policy Optimization Algorithms. Focus on §3, Eqs. 6–7 and Fig. 1: old/current probability ratio, minimum, clipping and the sign of the advantage. Read §5 and Algorithm 1 for repeated updates from one rollout batch. Use Block 5 ML02 for the worked calculation. The wider optimisation survey and benchmark results are optional. arXiv:1707.06347.

[R4] Rafailov et al. (2023; linked revision v3). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. Focus on §4, Eqs. 4–7: the optimum, reward reparameterisation, cancellation and DPO loss. Reproduce the corresponding steps developed in ML03. Appendix A.1 gives the full optimum derivation for optional deeper study; the paper's experiments are optional. arXiv:2305.18290v3.

[R5] Williams (1992). Simple statistical gradient-following algorithms for connectionist reinforcement learning. Machine Learning 8, 229–256. Original REINFORCE source; historical reading is optional. Use [R1] §§13.3–13.4 and Block 4 for preparation. DOI:10.1007/BF00992696.

Preparation sequence

1. Review all ten microlectures across Blocks 3–5 using the HTML and linked Instructor or AI version resources.
2. Rework the numerical examples on page 6 and explain the derivations.
3. Attempt the HTML self-check questions before viewing feedback; diagnose each error.
4. Use the focused readings to resolve gaps, then redo the affected calculation.

Scope boundary
Blocks 3–5 form the test portion. Explicitly optional material remains outside it: Block 3 returns/values supplement; Block 4 actor–critic supplement and variance-minimising-baseline aside; Block 5 per-token value/GAE supplement. Block 6 reasoning models, CoT, RLVR and GRPO are excluded. Earlier blocks provide prerequisite background. No new coding or report task is assigned by this guide.